



Module Title

Cybersecurity and Networking

Internship Assignment Submission

Kali Linux Introduction & Safe Lab Setup

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1.THEORY

1.1 Answer all 20 questions in your own words

1.What is Kali Linux?

Kali Linux is a Debian-based Linux operating system mainly used for cybersecurity, security testing, digital forensics, and training.

2. Why is Kali Linux used in cybersecurity?

Kali provides many tools that help security professionals test systems, find vulnerabilities, analyze networks, and perform authorized security testing.

3. What is a virtual machine?

A virtual machine is a computer system that runs inside other computer using virtualization software.

4. What is a hypervisor?

A hypervisor is software or technology that created and manages virtual machines.

5. What is VirtualBox?

VirtualBox is virtualization software that allows us to create and run virtual machines such as Kali Linux.

6. Explain Host OS and Guest OS.

The Host OS is the main operating system installed on the physical computer. The Guest OS is the operating system running inside the virtual machine.

7. Why is using Kali inside a VM useful for beginners?

It allows beginners to learn Kali without replacing their main operating system. It also provides isolation and makes it easier to reset the lab if something goes wrong.

8. What is a VM snapshot?

A VM snapshot saves the state of a virtual machine at a particular point in time so it can be restored later.

9. Why should students create snapshots?

Snapshots allow students to recover from mistakes, undo unwanted changes, and return to a clean Kali setup.

10. What does whoami display?

Whoami displays the username of the current logged-in user.

11. What does hostname display?

Hostname displays the name of a computer or virtual machine.

12. What does uname -a display?

Uname -a displays detailed information about the Linux kernel and system.

13. What does `/etc/os-release` contain?

`/etc/os-release` contains information about the operating system, such as its name and version.

14. What does `ip addr` show?

`ip addr` shows network interfaces and their IP addresses and configuration information.

15. What does `ip route` show?

`ip route` shows the system's routing information, including the default route used to reach other networks.

16. What is `sudo`?

`Sudo` allows an authorized user to run a command with elevated administrative privileges.

17. Why should elevated privileges be used carefully?

Elevated privileges provide greater control over the system. Using them unnecessarily can cause accidental changes or security problems.

18. What does `sudo apt update` do?

`Sudo apt update` refreshes the local package repository information. It does not automatically install upgrades.

19. Why should Kali Linux tools only be used on authorized systems?

Kali contains powerful security tools that can affect other systems. They should only be used where you have permission to avoid causing harm or breaking laws.

20. Explain the difference between ethical security testing and unauthorized access.

Ethical security testing is performed with permission, a defined scope, and legitimate security purpose. Unauthorized access means testing or accessing a system without permission, which can cause harm and may be illegal.

2. PRACTICAL LAB — Build and Explore Your Kali Linux Lab

2.1 SCREENSHOTS (from your own lab): Kali desktop · whoami · hostname · uname -a · cat /etc/os-release · ip addr · jayvik-cyber-lab directory · VirtualBox snapshot (Day11-Clean-Kali)

```
type help to get help:

(kali㉿kali)-[/home/kali]
PS> whoami
kali

(kali㉿kali)-[/home/kali]
PS> hostname
kali

(kali㉿kali)-[/home/kali]
PS> pwd
Path
/home/kali

(kali㉿kali)-[/home/kali]
PS> ls
1 3 5 7 9      Desktop  Downloads  My_Web_Project  Pictures  src  Videos
2 4 6 8 day10_lab Documents Music        Nessus-latest-debian10_amd64.deb  Public  Templates

(kali㉿kali)-[/home/kali]
PS> uname -a
Linux kali 6.8.11-amd64 #1 SMP PREEMPT_DYNAMIC Kali 6.8.11-1kali2 (2024-05-30) x86_64 GNU/Linux

(kali㉿kali)-[/home/kali]
PS> cat /etc/os-release
PRETTY_NAME="Kali GNU/Linux Rolling"
NAME="Kali GNU/Linux"
VERSION_ID="2024.3"
VERSION="2024.3"
VERSION_CODENAME=kali-rolling
ID=kali
ID_LIKE=debian
HOME_URL="https://www.kali.org/"
SUPPORT_URL="https://forums.kali.org/"
BUG_REPORT_URL="https://bugs.kali.org/"
ANSI_COLOR="1;31"
```

```
ANSI_COLOR="1;31"
```

```
(kali@kali)-[/home/kali]
```

```
PS> ip addr
```

```
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host nonprefixroute
```

```
(kali@kali)-[/home/kali]
```

```
PS> ping -c 4 google.com
```

```
/usr/bin/ping: google.com: Temporary failure in name resolution
```

```
(kali@kali)-[/home/kali]
```

```
PS> sudo apt update
```

```
[sudo] password for kali:
```

```
Ign:1 http://http.kali.org/kali kali-rolling InRelease
```

```
Ign:1 http://http.kali.org/kali kali-rolling InRelease
```

```
Ign:1 http://http.kali.org/kali kali-rolling InRelease
```

```
Err:1 http://http.kali.org/kali kali-rolling InRelease
```

```
Temporary failure resolving 'http.kali.org'
```

```
2251 packages can be upgraded. Run 'apt list --upgradable' to see them.
```

```
Warning: Failed to fetch http://http.kali.org/kali/dists/kali-rolling/InRelease Temporary failure resolving 'http.kali.org'
```

```
Warning: Some index files failed to download. They have been ignored, or old ones used instead.
```

```
(kali@kali)-[/home/kali]
```

```
PS> cd ~
```

```
(kali@kali)-[/home/kali]
```

```
PS> mkdir jayvik-cyber-lab
```

```
(kali@kali)-[/home/kali]
```

```
PS> cd jayvik-cyber-lab
```

```
(kali@kali)-[/home/kali/jayvik-cyber-lab]
```

```
PS> pwd
```

```
Path
```

```
/home/kali/jayvik-cyber-lab
```

```
(kali@kali)-[/home/kali/jayvik-cyber-lab]
```

```
PS> touch day11-notes.txt
```

```
(kali@kali)-[/home/kali/jayvik-cyber-lab]
```

```
PS> ls -l
```

```
total 0
```

```
-rw-rw-r-- 1 kali kali 0 Sep 18 10:06 day11-notes.txt
```

